

7

THE EFFECTS OF STOPS

One subject in geometrical optics, though very important from a practical standpoint, is frequently neglected because it does not directly concern the size, position, and sharpness of the image. This is the question of the *field of view*, which determines how much of the surface of a broad object can be seen through an optical system.

7.1 FIELD STOP AND APERTURE STOP

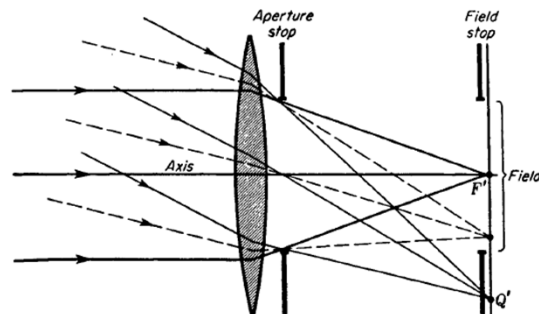


FIGURE 7A
The difference between a field stop and an aperture stop.

In Fig. 7A a single lens with two stops is shown forming the image of a distinct object. Three bundles of parallel rays from three different points on the object are shown brought to a focus in the focal plane of the lens. It can be seen from these bundles that the stop close to the lens limits the size of each bundle of rays, while the stop just in front of the focal plane limits the angle at which the incident bundles can get through to this plane. The first is called an *aperture stop*. It obviously determines the amount of light reaching any given point in the image and therefore controls the brightness of the latter. The second, or *field stop*, determines the extent of the object, or the field, that will be represented in the image.

7.2 ENTRANCE AND EXIT PUPILS

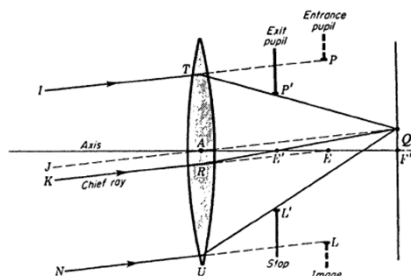


FIGURE 7B
How an aperture stop and its image can become the exit and entrance pupils, respectively, of a system.

A stop $P'E'L$ placed behind the lens as in Fig. 7B is in the image space and limits the image rays. By a graphical construction or by the lens formula, the image of this real stop, as formed by the lens, is found to lie at the position PEL shown by the broken lines. Since $P'E'L$ is inside the focal plane, its image PEL lies in the object space and is virtual and erect. It is called the *entrance pupil*, while the real aperture $P'E'L$ is, as we have seen, called the aperture stop. When it lies in the image space, as it does here, it becomes the *exit pupil*. (For a treatment of object and image spaces see Sec. 4.11.)

7.3 CHIEF RAY

Any ray in the object space that passes through the center of the entrance pupil is called a *chief ray*. Such a ray after refraction also passes through the center of the exit pupil. In any actual optical instrument the chief ray rarely passes through the center of any lens itself. The points E and E' at which the chief ray crosses the axis are known as the *entrance-pupil point* and the *exit-pupil point*. The former, as we shall see, is particularly important in determining the field of view.

7.4 FRONT STOP

In certain types of photographic lenses a stop is placed close to the lens, either before it (*front stop*) or behind it (*rear stop*). One of the functions of such a stop, as will be seen in Chap. 9, is to improve the quality of the image formed on the photographic film. With a front stop as shown in Fig. 7C its small size and its location in the object space make it the entrance pupil. Its image $P'E'L$ formed by the lens is in the image space and constitutes the exit pupil.

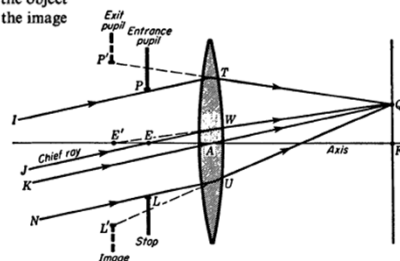


FIGURE 7C
A front stop and its image can become the respective entrance and exit pupils of a system.

While a certain stop of an optical system may limit the rays getting through the system from one object point, it may not be the aperture stop for other object points at different distances away along the axis. For example, in Fig. 7D a lens with a front stop is shown with an object point at M . For this point the periphery of the lens itself becomes the aperture stop, and since it limits the object rays, it is the entrance pupil. Its image, which is again the lens periphery, is also the exit pupil. The lens margin is therefore the aperture stop, the entrance pupil, and the exit pupil for the point M . If this object point were to lie to the left of Z , PEL would become the entrance pupil and the aperture stop and its image $P'E'L'$ the exit pupil.

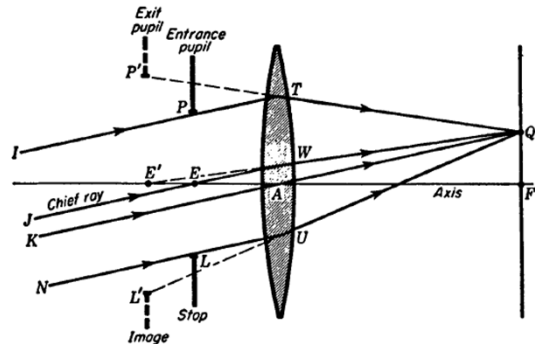


FIGURE 7C

A front stop and its image can become the respective entrance and exit pupils of a system.

7.5 STOP BETWEEN TWO LENSES

A common arrangement in photographic lenses is to have two separate lens elements with a variable stop, or iris diaphragm, between them. Figure 7E is a diagram representing such a combination, and in it the elements 1 and 2 are thin lenses while $P_0E_0L_0$ is the stop. By definition the entrance pupil of this system is the image of the stop formed by lens 1. This image is virtual, erect, and located at PEL . Similarly by definition the exit pupil of the entire system is the image of the stop formed by lens 2. This image, located at $P'E'L'$, is also virtual and erect. The entrance pupil PEL lies in the object space of lens 1, the stop $P_0E_0L_0$ lies in the image space of lens 1 as well as in the object space of lens 2, and the exit pupil $P'E'L'$ lies in the image space of lens 2.

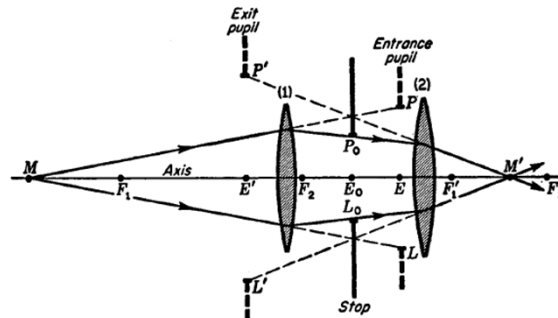


FIGURE 7E

Stop between two lenses. The entrance pupil of a system is in its object space, while the exit pupil is in its image space.

7.6 TWO LENSES WITH NO STOP

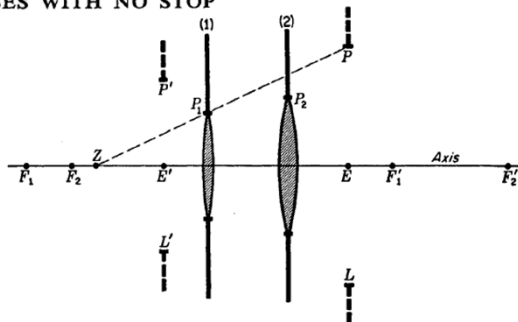


FIGURE 7G

The margin of any lens may be the aperture stop of the system.

The theory of stops is applicable not only where circular diaphragms are introduced into an optical system but to any system whatever, since actually the periphery of any lens in the system is a potential stop. In Fig. 7G two lenses, 1 and 2, are shown, along with their mutual images as possible stops. Assuming P_1 to be a stop in the object space, its image P' formed by lens 2 lies in the final image space. Looking upon P_2 as a stop in the image space, its image P formed by lens 1 lies in the first object space. There are therefore two possible entrance pupils, P_1 and P , in the object space of the combination of lenses, and two possible exit pupils, P_2 and P' , in the image space of the combination. For any axial point M lying to the left of Z , P_1 becomes the limiting stop and therefore the entrance pupil of the system. Its image P' becomes the exit pupil. If, on the other hand, M lies to the right of Z , P becomes the entrance pupil and P_2 the exit pupil.

7.7 DETERMINATION OF THE APERTURE STOP

the aperture stop of any system may change with a change in the object position. The general rule is that *the aperture stop of the system is determined by that stop or image of a stop which subtends the smallest angle as seen from the object point.*

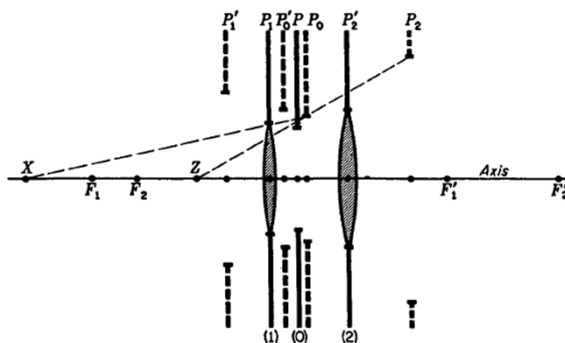


FIGURE 7H

A system composed of several elements has a number of possible stops and pupils.

7.8 FIELD OF VIEW

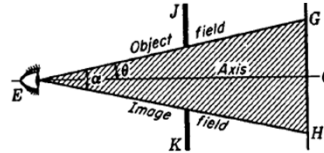


FIGURE 7I
Field of view through a window.

When one looks out at a landscape through a window, the field of view outside is limited by the size of the window and by the position of the observer. In Fig. 7I the eye of the observer is shown at E , the window opening at JK , and the observed field at GH . In this simple illustration the window is the field stop (Sec. 7.1). When the eye is moved closer to the window, the angular field α is widened, while when it is moved farther away, the field is narrowed. It is common practice with optical instruments to specify the field of view in terms of the angle α and to express this angle in degrees. The angle θ which the extreme rays entering the system make with the axis is called the *half-field angle* and limits the width of the object that can be seen. This object field includes the angle 2θ , and in this instance is the same as the image field of angular width α .

7.9 FIELD OF A PLANE MIRROR

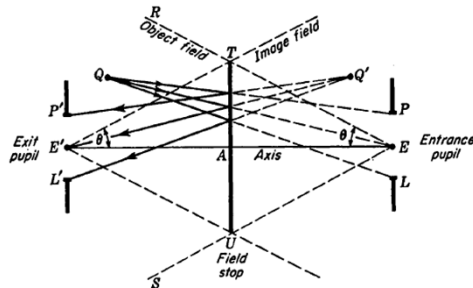


FIGURE 7J
Field of view looking in a plane mirror.

The field of view afforded by a plane mirror is very similar to that of a simple window. As shown in Fig. 7J, TU represents a plane mirror, and $P'E'L$ the pupil of the observer's eye, which here constitutes the exit pupil. The entrance pupil PEL is the virtual image of the eye pupil formed by the mirror and is located just as far behind the mirror as the actual pupil is in front of it. The chief rays $E'T$ and $E'U$ limit the field of view in image space, while the corresponding incident rays ER and ES define the field of view in object space. The latter show the limits of the field in which an object can be situated and still be visible to the eye. In this case also, although not in general, it subtends the same angle as the image field.

Since the limiting chief ray is directed toward the entrance pupil point E , the half-field angle θ is in general determined by the smallest angle subtended at E by any stop, or image of a stop, in the object space. The stop determined in this way is the field stop of the system. For a single mirror the field stop is the border of the mirror itself.

7.10 FIELD OF A CONVEX MIRROR

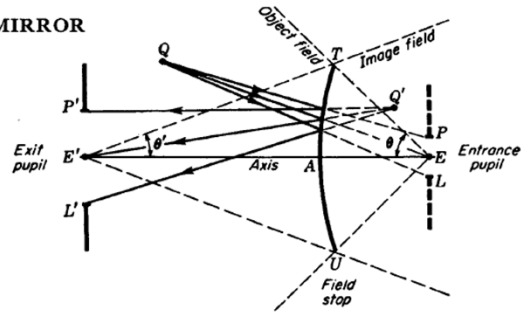


FIGURE 7K
Field of view looking in a convex mirror.

When the mirror has a curvature, the situation is little changed except that the object field and the image field no longer subtend the same angle ($\theta \neq \theta'$ in Fig. 7K). In this figure $P'E'L'$ represents the real pupil of an eye placed on the axis of a convex mirror TU . The mirror forms an image PEL of this exit pupil, and this is the entrance pupil which is now smaller. Following the same procedure as for a plane mirror, the lines limiting the image field and the object field have been drawn. Rays emanating from an object point Q toward P , E , and L of the entrance pupil are shown as reflected toward P' , E' , and L' in the exit pupil. When extended backward, these rays locate the virtual image Q' . The half-field angle θ is here larger than θ' , which determines the field of view to the eye. A similar but somewhat more complicated diagram can be drawn for the field of view of a concave mirror. This case will be left as an exercise

7.11 FIELD OF A POSITIVE LENS

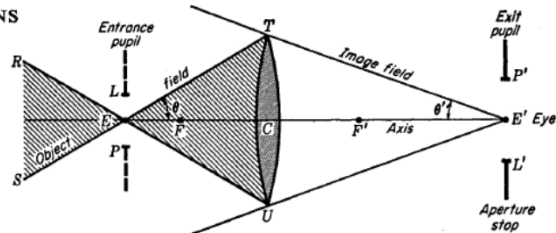


FIGURE 7L
Field of view looking in a converging lens.

The method of determining the half-field angles θ and θ' for a single converging lens is shown in Fig. 7L. The pupil of the eye, as an exit pupil, is situated on the right, and its real inverted image appears at the left. The chief rays through the entrance pupil point E which are incident at the periphery of the lens are refracted through the conjugate point E' .

The shaded areas, or rather cones, ETU and ERS mark the boundaries within which any object must lie in order to be seen in the image field. The field stop in this case is the lens TU itself, since it determines the half-field angle subtended at the entrance pupil point. If the eye, and therefore the exit pupil, is moved closer to the lens, thereby increasing the image-field angle θ' , the inverted entrance pupil moves to the left, causing a lengthening of the object-field cone ETU .

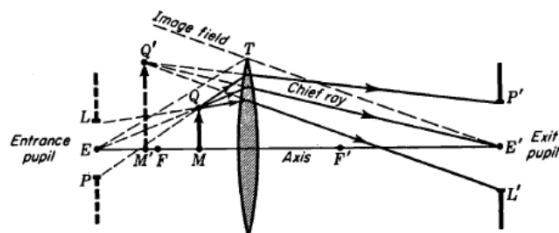


FIGURE 7M
Image formation within the field of a converging-lens system.

The same lens has been redrawn in Fig. 7M, where an object QM is shown in a position inside the primary focal point. Through each of the three points P , E , and L , rays are drawn from Q to the lens. From there the refracted rays are directed through the corresponding points P' , E' , and L' on the exit pupil. By extending them backward to their common intersection, the virtual image is located at Q' . The oblique-ray or parallel-ray methods of construction (not shown) may be used to confirm this position of the image. It will be noted that if objects are to be placed near the entrance pupil point E , they must be very small; otherwise only a part of them will be visible to an eye placed at E' . The student will find it instructive to select object points that lie outside the object field and to trace graphically the rays from them through the lens. It will be found that invariably they miss the exit pupil.

When a converging lens is used as a magnifier, the eye should be placed close to the lens, since this widens the image-field angle and extends the object field so that the position of the object is less critical.